Shikhar Mishra

Noida, India | +91-7618091967 | shikharmishra.nu@gmail.com | linkedin.com/in/shikhar-mishra-17b5a1231 | github.com/shikharmishra1

Professional Summary

AI Engineer specializing in production-grade Retrieval-Augmented Generation (RAG), agentic AI, semantic search, document intelligence, and NLP systems. Delivered 93% answer accuracy, 96% embedding recall, 2x active-user growth, and 120% longer sessions using Python, FastAPI, PyTorch, LangChain, Apache Solr, and scalable microservices.

Experience

1Ansah Technologies India
AI Engineer *July 2024 – July 2026*

- Architected tool-calling ReAct agents and RAG pipelines, improving answer accuracy to **93%** through multi-step reasoning and tool orchestration.
- Built enterprise document-intelligence systems across **10,000+ documents** using BM25, dense retrieval, SentenceTransformers, cross-encoder reranking, and semantic caching.
- Developed NLP search and personalization pipelines that increased active users by **2x** and average session duration by **120%**.
- Engineered FastAPI microservices integrated with Apache Solr, MongoDB, MinIO, Keycloak, and LLM inference infrastructure.
- Designed Learning-to-Rank pipelines using RankLib, behavioral signals, click models, and semantic embeddings.
- Fine-tuned a domain-aware SentenceTransformer embedding model for aviation search, achieving **96% recall**.

Projects

TurboParse | *TypeScript, LightGBM, ONNX Runtime Web, WebAssembly, OpenCV* | Live Demo

- Built a privacy-first, browser-native document intelligence engine that parses documents entirely on-device, supporting offline and air-gapped deployments without external APIs or server-side inference.
- Trained a **LightGBM document-layout classification model** on DocLayNet, exported it to the **ONNX** format, and integrated it with **ONNX Runtime Web** for low-latency CPU inference directly inside the browser.
- Achieved **91.34% classification accuracy** and **86.32% macro F1** on DocLayNet while processing approximately **100 pages per second** without a GPU.
- Developed a document-processing pipeline using OpenCV, WebAssembly, PDF text extraction, geometric layout analysis, and parallel page processing to identify tables, text, figures, formulas, headings, and list elements.
- Designed a layout-aware intermediate representation that preserves bounding boxes, reading order, document structure, and element metadata for generating structured HTML, Markdown, search indexes, and RAG-ready content.
- Implemented adaptive WebP/PNG encoding, bounded concurrent processing, browser-compatible fallbacks, and memory-efficient serialization for large-document processing on desktop and mobile devices.

AI Power System Analysis | *React, FastAPI, LangChain, PyTorch, MongoDB*| Live Demo

- Built a full-stack AI platform for electrical-network analysis spanning 14 to 70,000+ buses with Cytoscape.js and MongoDB GridFS.
- Developed a LangChain ReAct agent with **60+ tools** for topology, power flow, contingency, OPF, and SCADA queries; achieved RAGAS ToolCallAccuracy and Faithfulness of **1.00**.
- Integrated a YOLO schematic-to-model pipeline, trained a PyTorch graph neural network for state prediction, and packaged 100+ MATPOWER benchmark cases.

Nirguna Scripting Language | *TypeScript, Compiler Design, VS Code Extension*| Live Demo

- Built a Unicode-native Devanagari programming language with a custom lexer, recursive-descent parser, type-safe abstract syntax tree, tree-walking interpreter, and REPL.
- Implemented lexical scoping, first-class functions, closures, objects, arrays, localized errors, and a VS Code syntax-highlighting extension.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL
Generative AI / NLP: LLMs, Retrieval-Augmented Generation (RAG), Agentic AI, ReAct, fine-tuning, embeddings, reranking, RAGAS
Machine Learning: PyTorch, Hugging Face Transformers, SentenceTransformers, cross-encoders, graph neural networks, YOLO
Search / Backend: Apache Solr, BM25, dense vector search, hybrid search, Learning-to-Rank, RankLib, FastAPI, MongoDB, MinIO
Infrastructure / Tools: Docker, Docker Compose, Azure, Keycloak, React, Git, OCR pipelines

Education

University of Lucknow Lucknow, India
Master of Computer Applications (MCA) *2022 – 2024*
Sherwood College of Professional Management Lucknow, India
Bachelor of Computer Applications (BCA) *2019 – 2022*